Practice Assignment 2

Logistic Regression

Discussion on Friday, May 8th at 4 PM

1) This problem will use the **USTempLocation.csv** file on GitHub. In this data set, the average low temperature in January is given along with latitude and longitude of various cities in the United States. Additionally, the cities region in the US (Northeast, Midwest, South, or West) and whether the city is over 1 million in population or not are given.

   a. Read in the dataset and load the **math3150package** library. Then change the **CitySize** variable to be a factor with the first level "Under1M" and the second level "Over1M". This can be done using
   ```
   city_data <- mutate(
     city_data,
     CitySize = factor(CitySize, levels = c("Under1M", "Over1M"))
   )
   ```
   where city_data is the name of the data frame.

   b. Fit a logistic regression model predicting the size of a city using JanTemp and Region. Don't forget to put **family = binomial()**. Write down or type the logistic regression model here.

   c. Are any of the variables significant in this model? If so, which ones and how do you know?

   d. Which region is being treated as the baseline here? Explain how you know.

   e. Interpret the slope of JanTemp in the model.

   f. Interpret the slope of RegionSouth in the model.

   g. Find the predicted probability of the city having a population over 1 million if it is in the Northeast region with a January low temperature of 20 degrees. Putting **type = "response"** in the **predict()** function will be helpful here.

   h. Produce a table with all the actual and all the predicted values from this model using 0.50 as the probability cutoff for predicted the city has over 1 million people. What percent of cities were classified correctly? Does it seem like the model is doing a good job at predicting city size?

   i. Use the **logistic_plots()** function in **math3150package** to make plots for the model. Based on these plots, comment on the diagnostics for this model.

2) Three airlines serve a small town in Ohio. Airline A has 45% of all scheduled flights, airline B has 38% and airline C has the remaining 17%. Their on-time rates are 82%, 73%, and 51%, respectively. A flight just left on time. What is the probability that it was a flight of airline A?